Arbitrarily clustered hot spots on bridgeless quantum graphs

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Abstract

We construct finite compact metric multigraphs with standard Kirchhoff conditions, diameter one, and a simple first positive Laplacian eigenvalue, whose complete set of global maxima and minima has arbitrarily small diameter. Every edge belongs to a cycle. This proves Conjecture 8.9 of Kennedy and Rohleder on the hot spots of quantum graphs in its stated no-bridge sense. The construction combines an explicit specialization of their split-tree family with the known operation of uniformly doubling edges. For an integer $m \geq 4$, the first positive eigenvalue is $\left[\arctan((2m+1)^{-1/2})/h_m\right]^2$, where $h_m = \pi^{-1}\arctan(m^{-1/2})$, and the complete extrema set has diameter $6h_m$. We prove the ordering and simplicity in the full operator domain and control distances between interior points of the doubled edges. The result concerns homogeneous network diffusion; articulation vertices and growing total length and degree are allowed.

Keywords: metric graphs; Kirchhoff Laplacian; hot spots; network diffusion; spectral geometry.

1 The hot-spot distance question

Let Γ be a finite connected compact metric multigraph. Its edges are intervals of strictly positive finite lengths, and H_Γ acts as $-\mathrm{d}^2/\mathrm{d}x^2$. The domain consists of edgewise H^2 functions that are continuous at each vertex and satisfy

$$\sum_{e\sim v} \partial_{\nu_e} f(v) = 0. \tag{1.1}$$

Here ∂_{ν_e} points from the vertex into the incident edge. At a vertex of degree one this is a Neumann condition. The Hilbert space is the unweighted space $L^2(\Gamma) = \bigoplus_e L^2(0, L_e)$. The associated energy form is

$$q_\Gamma(f) = \sum_e \int_0^{L_e} |f_e'(x)|^2 \,\mathrm{d}x$$

on continuous edgewise H^1 functions. Thus the spectrum is discrete, $0 = \mu_1(\Gamma) < \mu_2(\Gamma) \leq \mu_3(\Gamma) \leq \cdots$, with a simple constant zero mode.

When $\mu_2(\Gamma)$ is simple, take any nonzero real eigenfunction ψ_2 and define

$$\mathcal{M}(\Gamma) = \arg\max_\Gamma \psi_2 \;\cup\; \arg\min_\Gamma \psi_2. \tag{1.2}$$

This set is unchanged by rescaling or reversing the sign of ψ_2. Both extrema are global over the whole metric graph, including edge interiors and all ties. All distances below are shortest-path distances within the graph.

Kennedy and Rohleder proved that $\operatorname{diam}\mathcal{M}/\operatorname{diam}\Gamma$ can be arbitrarily small on trees [2, Proposition 4.11 and Example 6.2]. Their Conjecture 8.9 asks for the same conclusion on

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doubly connected metric graphs. Their definition in Section 4.2 means that every point lies on a cycle; for the finite connected multigraphs considered here this is the absence of bridges. It permits articulation vertices. The following explicit construction establishes that conjecture.

Theorem 1. *For every $\varepsilon > 0$ there exists a finite connected compact standard-Kirchhoff metric multigraph Γ such that every edge lies on a cycle,*

$$\operatorname{diam}\Gamma = 1, \qquad \dim\ker(H_\Gamma - \mu_2(\Gamma)) = 1, \qquad \operatorname{diam}\mathcal{M}(\Gamma) < \varepsilon.$$

More precisely, for each integer $m \geq 4$, set

$$h_m = \frac{\arctan(m^{-1/2})}{\pi}, \qquad \theta_m = \arctan((2m+1)^{-1/2}), \qquad k_m = \frac{\theta_m}{h_m}. \tag{1.3}$$

There is a graph Γ_m for which

$$\mu_2(\Gamma_m) = k_m^2 < \pi^2, \qquad \operatorname{diam}\Gamma_m = 1, \qquad \operatorname{diam}\mathcal{M}(\Gamma_m) = 6h_m, \tag{1.4}$$

and $\mathcal{M}(\Gamma_m)$ consists of exactly $2m$ vertices. The choice

$$m = \max\left\{4, \left\lceil \frac{36}{\pi^2\varepsilon^2} \right\rceil\right\} \tag{1.5}$$

gives the required strict inequality.

Our comparison tree is a specialization of the published split-tree construction [2, Example 6.2]. Uniform replacement of every edge by two equal-length parallel edges is an established operation, discussed in [1, Section 4.3]. The contribution here is the explicit no-bridge consequence, with full spectral ordering, the complete extrema set, and the continuum metric diameter proved below. We make no claim of a new doubling method or exhaustive historical priority. A bounded primary-literature check through October 2026 identified no earlier equivalent resolution.

2 An explicit comparison tree

Fix $m \geq 4$, abbreviate $h = h_m$, $\theta = \theta_m$, and $k = k_m$, and let T_m be the following tree. At a central vertex o, attach two long arms of length $a = 1/2$, ending at ℓ_1, ℓ_2. Attach also two stems of length h, ending at hubs p, n. Each hub has m leaf edges of length $b = 2h$, with tips $p_1, \ldots, p_m$ and $n_1, \ldots, n_m$. Use standard Kirchhoff conditions throughout, hence Neumann conditions at all tips. Define Γ_m by replacing *every* edge of T_m by two parallel edges of the same original length; see Figure 1.

The parameter choice ensures

$$0 < h \leq \frac{\arctan(1/2)}{\pi} < \frac{1}{6}, \qquad b = 2h < \frac{1}{3}, \qquad h + b = 3h < \frac{1}{2}. \tag{2.1}$$

In the source construction the unlengthened fan leaf has length $\pi^{-1}\arctan(\cot(\pi h)/m)$, which equals h at our choice of h. Thus $b = 2h$ corresponds to lengthening each fan leaf by h. The proof below orders the resulting spectrum directly; it does not require an unspecified small perturbation.

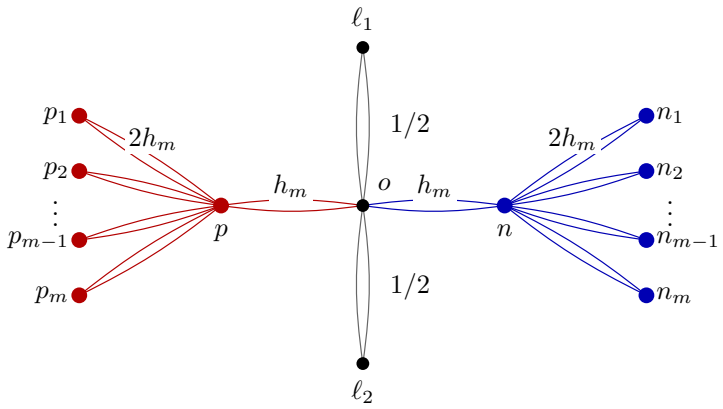


Figure 1: Schematic of Γ_m, with representative fan edges shown. There are m distinct tips at each fan hub, and every displayed edge has two equal-length copies. The two long pairs have length $1/2$ per copy, the stems length h_m, and the fan edges length $2h_m$. The drawing is not to metric scale. Red and blue tips are respectively all global maxima and all global minima of the mode in Section 5; both long pairs have zero mode value. Deleting one copy of each edge gives the comparison tree T_m.

3 The first positive tree eigenvalue

We first retain the full leaf-permutation decomposition. At either hub, write the m leaf functions as their common mean plus $m-1$ zero-sum components. Continuity forces the zero-sum components to vanish at the hub, while their endpoint fluxes cancel in the sum. These components therefore give $m-1$ Dirichlet–Neumann interval sectors of length b per fan. Their wave numbers are

$$\frac{(2r+1)\pi}{2b} = \frac{(2r+1)\pi}{4h}, \qquad r = 0, 1, 2, \dots. \tag{3.1}$$

Their first wave number exceeds π, since $h < 1/6 < 1/4$. The complementary invariant subspace has identical functions on all leaves at each hub. This is a decomposition of the whole operator, rather than an imposed equal-leaf ansatz.

3.1 A single fan with a Dirichlet root

Put a Dirichlet condition at the root o of one fan, and use coordinates $x \in [0, h]$ from root to hub and $y \in [0, b]$ from hub to a leaf tip. At a positive wave number t, its equal-leaf mode has the form

$$w_s(x) = S\sin(tx), \qquad w_j(y) = C\cos(t(b-y)).$$

Continuity and Kirchhoff flux at the hub give

$$C\cos(tb) = S\sin(th), \qquad mC\sin(tb) = S\cos(th). \tag{3.2}$$

The characteristic equation, written without division by a trigonometric factor, is

$$F(t) := \cos(th)\cos(tb) - m\sin(th)\sin(tb) = 0. \tag{3.3}$$

For $0 < t \le \pi$, both th and tb belong to $(0, \pi/2)$. The function $m\tan(th)\tan(tb)$ is strictly increasing from zero and at $t = \pi$ equals

$$m\frac{1}{\sqrt{m}}\frac{2/\sqrt{m}}{1-1/m} = \frac{2m}{m-1} > 1.$$

Hence the fan has exactly one Dirichlet-root wave number below π. Writing $z = \tan(th)$ and using $b = 2h$, its equation becomes

$$\frac{2mz^2}{1-z^2} = 1, \qquad z^2 = \frac{1}{2m+1}.$$

The root is precisely $t = k$, is simple, and satisfies $k < \pi$. Together with (3.1), this also shows that the full rooted fan has no lower positive eigenvalue. The two rooted long arms are Dirichlet–Neumann intervals of length $1/2$, whose first wave number is π.

3.2 Excluding lower full-tree modes and proving simplicity

Suppose $0 < t < k$ and let c denote an eigenfunction's value at o. There are no leaf zero-sum modes in this range. On each long arm the outgoing derivative at o is

$$ct\tan(t/2).$$

On each equal-leaf fan, solving the two hub equations gives outgoing root derivative

$$ct\,\frac{\tan(th) + m\tan(tb)}{1 - m\tan(th)\tan(tb)}.$$

Both coefficients multiplying c are strictly positive throughout $0 < t < k$. Thus the central Kirchhoff condition is impossible when $c \neq 0$. When $c = 0$, all four root-Dirichlet branches are invertible in this range, so the function vanishes. This excludes every positive eigenvalue below k^2.

At $t = k$, allow a nonzero root value and write the stem as $c\cos(kx) + S\sin(kx)$. The exact hub equations are

$$C\cos(kb) - S\sin(kh) = c\cos(kh), \tag{3.4}$$
$$mC\sin(kb) - S\cos(kh) = -c\sin(kh). \tag{3.5}$$

Because $F(k) = 0$, the left side of (3.5) is $\cot(kh)$ times the left side of (3.4). Compatibility therefore requires

$$-c\sin(kh) = \frac{c\cos^2(kh)}{\sin(kh)},$$

and hence $c = 0$. No singular expression is used outside its stated range, where $\sin(kh) > 0$.

The long arms consequently vanish, since $k < \pi$. Each fan has one remaining amplitude S_+ or S_-, and the central flux equation is

$$k(S_+ + S_-) = 0.$$

It leaves exactly one nonzero eigendirection. We have proved

$$\mu_2(T_m) = k^2 < \pi^2, \qquad \dim\ker(H_{T_m} - k^2) = 1. \tag{3.6}$$

This proof includes all sectors and orders the lowest positive eigenvalue in the full standard domain.

4 Uniform doubling and the full operator

For clarity we state the complete decomposition behind the known doubling operation. It is essential here to retain the additional Dirichlet sectors.

Lemma 1. *Let T be a finite compact standard-Kirchhoff metric graph and let G replace every edge e of length L_e by two equal parallel copies. Then*

$$H_G \;\simeq\; H_T \,\oplus\, \bigoplus_{e\in E(T)} H_{\mathrm{DD}}(0,L_e). \tag{4.1}$$

The direct sum includes all eigenvalues and their multiplicities.

Proof. Align the coordinates of each pair and define

$$g_e = \frac{f_{e,1}+f_{e,2}}{\sqrt{2}}, \qquad q_e = \frac{f_{e,1}-f_{e,2}}{\sqrt{2}}.$$

This transformation is unitary in the edgewise L^2 spaces. Continuity of f makes q_e zero at both endpoints. All incident endpoint values of g are equal, being $\sqrt{2}$ times the common physical vertex value. The sum of the two copied-edge derivatives at a vertex is $\sqrt{2}$ times the corresponding derivative of g_e; the q_e derivatives cancel. Since every original edge is doubled, the full vertex equation reduces exactly to the Kirchhoff equation for g on T.

Conversely, any $g \in D(H_T)$ and independent $q_e \in H^2(0,L_e)\cap H^1_0(0,L_e)$ reconstruct $f_{e,1} = (g_e+q_e)/\sqrt{2}$ and $f_{e,2} = (g_e-q_e)/\sqrt{2}$ in $D(H_G)$. The differential actions commute with this transformation. This two-sided domain map proves (4.1). □

Each added interval has eigenvalues $(r\pi/L_e)^2$, $r = 1,2,\ldots$. For T_m, all $L_e \le 1/2$. Thus every added eigenvalue is at least $4\pi^2$, strictly above the simple tree eigenvalue $k^2 < \pi^2$. Equation (4.1) therefore gives

$$\mu_2(\Gamma_m) = k^2, \qquad \dim\ker(H_{\Gamma_m} - k^2) = 1. \tag{4.2}$$

Its eigenfunction has equal profiles on paired copies. Uniform doubling is used at all vertices; doubling only a selection of edges would in general change the symmetric vertex equations and would not justify this conclusion.

5 Every extremum and every metric distance

Choose the tree eigenfunction with $S_+ = 1$ and $S_- = -1$. The long arms are identically zero. On the positive stem and on each positive fan edge its profiles are

$$\psi_s(x) = \sin(kx), \qquad \psi_j(y) = C_m\cos(k(2h-y)), \qquad C_m = \frac{\sin\theta}{\cos(2\theta)} = \frac{\sqrt{m+1}}{\sqrt{2}\,m} > 0. \tag{5.1}$$

The negative fan has the opposite profiles. Indeed,

$$\sin\theta = \frac{1}{\sqrt{2m+2}}, \qquad \cos(2\theta) = \frac{m}{m+1}, \qquad \sin(2\theta) = \frac{\sqrt{2m+1}}{m+1},$$

which verify both equations in (3.2). Use these same unnormalized profiles on both copies in Γ_m.

Because $0 < \theta < \pi/4$, the positive stem is strictly increasing away from o, and every positive fan edge is strictly increasing up to its tip: its derivative is positive on $0 \le y < 2h$. The hub value is $C_m\cos(2\theta) < C_m$. The negative fan is strictly decreasing in the same orientations. All other points have values strictly between the global extremes. Consequently

$$\mathcal{M}(\Gamma_m) = \{p_1,\ldots,p_m,n_1,\ldots,n_m\}. \tag{5.2}$$

In particular, the equal values on the long arms do not create global extrema, and none of the copied edge interiors is omitted.

On T_m, tips at the same hub are distance $4h$ apart, and tips at opposite hubs are distance $6h$ apart. The long tips ℓ_1, ℓ_2 are distance one apart; any long tip and any fan tip are distance $1/2 + 3h < 1$ apart. A geodesic in a finite tree extends, if necessary, to a path between leaves. These leaf distances therefore prove the full continuum diameter $\operatorname{diam} T_m = 1$.

Folding every parallel pair onto its tree edge defines a map $P : \Gamma_m \to T_m$ that does not increase path lengths. Hence

$$\operatorname{dist}_{T_m}(Px, Py) \le \operatorname{dist}_{\Gamma_m}(x, y).$$

For vertices, the tree geodesic lifts to Γ_m, giving equality. The same is true for points that project to distinct original edges: the unique tree path lifts with the prescribed copy at either end. Points on a single common copy also have their tree distance. Only points in opposite copies of the *same* original edge need a separate estimate.

For that last case, use coordinates $s, t \in [0, L]$ on aligned copies. A route through either endpoint has length $s + t$ or $2L - s - t$, respectively. There is no external path between these endpoints avoiding the pair, since deleting the original edge disconnects the tree. Thus their exact distance is

$$\min\{s + t,\, 2L - s - t\} \le L \le \frac{1}{2}. \tag{5.3}$$

These cases exhaust all pairs of points in the doubled graph. Every distance is at most one, and the two long tips still realize one. Vertex distances, including all distances in (5.2), are unchanged by doubling. Therefore

$$\operatorname{diam} \Gamma_m = 1, \qquad \operatorname{diam} \mathcal{M}(\Gamma_m) = 6h.$$

Each copied edge belongs to the two-edge cycle formed with its mate. Thus Γ_m has no bridge and every point lies on a cycle. Finally, $\arctan t < t$ for $t > 0$ and (1.5) imply

$$6h_m = \frac{6}{\pi} \arctan(m^{-1/2}) < \frac{6}{\pi\sqrt{m}} \le \varepsilon.$$

The chosen m is finite for every positive ε. This completes the proof of Theorem 1.

6 Network diffusion and the scope of the construction

For a uniform physical diffusivity $D > 0$, the heat equation on these networks is

$$\partial_t u = -D H_\Gamma u.$$

Kirchhoff flux conserves the integral of u, and the constant mode is equilibrium. If the initial temperature has a nonzero projection onto the simple first positive mode, its slowest nonconstant component is

$$A e^{-D\mu_2 t} \psi_2.$$

For that component the entire hottest/coldest set has the spatial diameter calculated above. A sufficiently large constant background makes this sign-changing temperature deviation compatible with positive absolute temperature. The statement about $\mathcal{M}$ is exactly about the eigenmode; at finite times higher modes may split the symmetry ties.

Taking the diameter as one length unit, Theorem 1 excludes a positive universal lower bound on the fraction of network diameter spanned by the complete extrema set of a simple slowest mode, even when all edges are cyclic. Restoring a length scale L_0 multiplies all distances by L_0 and sends the eigenvalue to k_m^2/L_0^2. Thus $D\mu_2$ has units of inverse time.

The construction has $4m + 8$ edges after doubling and total length

$$|\Gamma_m| = 2 + 4h_m + 8mh_m. \tag{6.1}$$

The degree of each fan hub is $2m + 2$. Both total length and hub degree grow along the family. Articulation vertices remain. We therefore establish the precise no-bridge conjecture, without adding fixed-volume, uniformly bounded-degree, vertex-biconnected, Euclidean-domain, or experimental conclusions.

All spectral, extremal, and metric assertions in the proof are exact for every integer $m \geq 4$. The accompanying reproducibility files provide the analytic argument and diagnostic programs based on full endpoint matching, exact algebraic identities, and optimization over interior edge-pair coordinates. Floating-point roots, ranks, and metric samples are diagnostics rather than interval certificates, and are not used to prove the quantified existence statement.

Use of generative AI tools

OpenAI Codex was used for literature screening, development of the construction and analytic arguments, implementation of diagnostic calculations, internal checking, and drafting the text and the diagram. These uses include AI-generated scientific text and proof development throughout the article. Internal AI checks are not external peer review and may share errors; literature coverage is bounded. Human scientific review and author approval of this draft remain pending.

Data and code availability

The reproducibility supplement contains the explicit construction, the exact analytic argument, input records, diagnostic code, and environment and regeneration instructions. It contains no experimental data. The proof does not depend on a numerical enclosure or spatial truncation.

References

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